

IBM EMPLOYEE ATTRITION ANALYSIS

Professional End-to-End Data Analytics Project Report

Excel • Power Query • Power BI • MySQL

Project component	Implementation
Source / cleaned data	Excel workbook — raw source + 43-column analytical table; Database source: Kaggle — IBM HR Analytics Employee Attrition & Performance (https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset)
Data preparation	Excel Power Query
Data modeling & measures	Power BI
Dashboard	One-page executive HR attrition dashboard
Validation & analysis	MySQL 8.0+ single-table SQL
Analytical focus	Workforce size, attrition, role/department risk, demographics, tenure, compensation, satisfaction and work conditions

Prepared from the supplied Excel workbook, PBIX, SQL script and screenshot evidence.

1. Executive Summary

This project develops an end-to-end HR analytics solution using an employee-level attrition dataset. The cleaned analytical table contains 1,470 employee records and 43 columns. The source worksheet contains 35 original fields, while the cleaned table adds business-ready classifications such as Age_Group, Income_Category, Company_Tenure, Experience_Group, Employee_Status, Distance_Category, Performance_Category and Attrition_Risk.

The workforce contains 1,470 employees: 1,233 active employees and 237 employees who left. The overall attrition rate is 16.1%, while the active-employee rate is 83.9%. Average age is 36.9 years, average company tenure is 7.0 years, average monthly income is \$6.5K, and average overall satisfaction is 2.73.

Executive KPI Snapshot

KPI	Value	Business meaning
Total Employees	1,470	Employees represented in the analytical population
Employees Left	237	Employees with Attrition = Yes
Active Employees	1,233	Employees with Attrition = No
Attrition Rate	16.1%	Overall employee exit rate
Active Employee Rate	83.9%	Share retained in the dataset
Average Age	36.9 years	Average employee age
Average Tenure	7.0 years	Average years at company
Average Salary	\$6.5K	Average monthly income
Average Satisfaction	2.73 / 4	Composite satisfaction measure

Headline Findings

- Attrition is concentrated among younger employees: the 18–25 group has a 35.8% attrition rate.
- Early-career employees are the strongest retention concern: Fresher (0–2) employees have a 43.9% attrition rate.
- Overtime is strongly associated with attrition: 30.5% for employees working overtime versus 10.4% without overtime.
- Sales has the highest department attrition rate at 20.6%.
- Sales Representative is the highest-risk job role by attrition rate at 39.8%.
- High Risk employees show 37.2% attrition compared with 10.4% for Low Risk employees.

2. Report Structure

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3. Project Definition & Business Questions

The project is designed to help HR and management understand employee attrition patterns and identify the workforce segments that require retention attention. The analytical scope combines organizational structure, employee demographics, career stage, compensation, satisfaction, workload, travel and a derived multi-factor attrition-risk classification.

Business Questions

- How large is the workforce and what is the overall attrition rate?
- Which departments and job roles have the highest attrition?
- How does attrition vary by age, gender and marital status?
- Does attrition concentrate in early career or short company-tenure groups?
- How does income category relate to employee attrition?
- How strongly is overtime associated with employee exits?
- How do satisfaction and work-life balance differ across attrition-risk groups?
- Does business travel correspond with different attrition rates?
- Which employees are classified as High, Moderate and Low Risk under the project's rule?
- Can the dashboard's headline KPIs be independently reproduced in SQL?

Project Objectives

- Create a clean and standardized employee-level analytical table.
- Convert source attributes into useful HR segmentation fields.
- Build a Power BI reporting layer for executive and exploratory analysis.
- Validate core HR KPIs and group-level calculations through MySQL.
- Translate analytical patterns into practical retention recommendations.

4. Dataset & Workbook Structure

Database Source: Kaggle — IBM HR Analytics Employee Attrition & Performance.

<https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>

The supplied Excel workbook contains two main employee-data worksheets. The raw worksheet, named 'Master_Data', contains 1,470 rows and 35 columns. The cleaned 'Cleaned_Data' contains the same 1,470 employee records and expands the field set to 43 columns.

Workbook object	Rows × columns	Purpose
Master_Data	1,470 × 35	Original source-stage employee attributes
Cleaned_Data	1,470 × 43	Cleaned and engineered analytical table

Field Families

Family	Examples	Analytical use
Identifiers	Employee_ID	Employee-level grain and validation
Attrition	Attrition, Attrition_Flag, Employee_Status	Retention KPI and status
Demographics	Age, Gender, Marital_Status, Education_Field	Workforce segmentation
Organization	Department, Job_Role, Job_Level	Role and department analysis
Compensation	Monthly_Income, Daily_Rate, Hourly_Rate, Monthly_Rate	Pay analysis
Career / tenure	Total_Working_Years, Years_At_Company, Years_In_Current_Role	Career-stage analysis
Satisfaction	Environment, Job, Relationship, Work_Life_Balance, Overall_Satisfaction	Employee experience
Work conditions	Over_Time, Business_Travel, Distance_From_Home	Potential attrition drivers
Derived segmentation	Age_Group, Income_Category, Experience_Group, Attrition_Risk	Dashboard-ready categories

5. End-to-End Analytical Workflow

Stage	Tool	Activities	Deliverable
1	Excel	Source employee dataset	Raw worksheet
2	Power Query	Type conversion, standardization, derived categories, validation and duplicate removal	Cleaned_Data /cleaned dataset
3	Power BI	Semantic model and measure layer	Analytical model
4	Power BI	KPI cards, charts, table, treemap and slicers	Executive dashboard
5	MySQL	Structure checks, KPI calculations, group analysis and multi-factor risk analysis	Independent validation layer
6	Business analysis	Interpretation and retention actions	Insights & recommendations

Traceability

- Power Query creates the standardized fields used by the dashboard.
- Power BI measures centralize recurring KPI logic.
- SQL calculates the same workforce and attrition concepts independently.
- The supplied screenshots preserve both the dashboard and the Power Query transformation interface.

6. Excel & Power Query Preparation

The Power Query evidence shows a single query named Cleaned_Table. The transformation chain starts from the source, standardizes data types and columns, creates business groupings, adds employee status and attrition logic, and finishes by removing duplicates based on Employee_ID.

Cleaned Output

Output characteristic	Observed result
Records	1,470
Columns	43
Employee_ID unique	Yes
Missing values in Cleaned_Data	None observed
Attrition flag consistency	Consistent with Attrition
Duplicate Employee_ID after cleaning	0

Major Transformation Themes

- Age is converted into Age_Group.
- Monthly income is converted into Income_Range and Income_Category.
- Years at company is converted into Company_Tenure.
- Total working years is converted into Experience_Group.
- Employee status is derived as Active Employee / Ex-Employee.
- Attrition_Flag is created from Attrition.
- Distance is categorized into Near, Moderate and Far.
- Performance is categorized into Good and Excellent.
- Overall satisfaction is derived from the satisfaction components.
- Attrition_Risk is derived from overtime, satisfaction, promotion gap and distance.
- Text values are cleaned with replacement and trimming steps.
- Employee_ID duplicates are removed at the end of the query.

7. Power Query Applied Steps

Step	Applied Step observed in evidence
1	Source
2	Changed Type
3	Reordered Columns
4	Added Age Group
5	Changed Age group Data type
6	Replaced value in age group
7	Added Income Range
8	Added Income Category
9	Added Company_Tenure
10	Added Experience_Group
11	Added Employee_Status
12	Added Attrition_Flag
13	Added Distance_Category
14	Added Performance_Category
15	Renamed Columns
16	Added Overall_Satisfaction
17	Added Attrition_Risk
18	Replaced Value
19	Changed DataType
20	Trimmed Text
21	Removed Constant value Column
22	Reordered Columns1
23	Removed Duplicates

The supplied Power Query screenshot shows the complete Applied Steps list and the final table preview. The final step is Removed Duplicates, with the formula bar showing Table.Distinct applied to Employee_ID.

The screenshot displays the Power Query interface. On the left, the 'Queries' pane shows 'Cleaned_Table'. The main area shows a table with 43 columns: Employee_ID, Employee_Status, Attrition_Flag, Attrition, Age, Age_Group, Gender, and others. The status bar at the bottom indicates '43 COLUMNS, 999+ ROWS'. On the right, the 'Query Settings' pane shows the 'APPLIED STEPS' list, which includes 23 steps: Source, Changed Type, Reordered Columns, Added Age Group, Changed Age group Data type, Replaced value in age group, Added Income Range, Added Income Category, Added Company_Tenure, Added Experience_Group, Added Employee_Status, Added Attrition_Flag, Added Distance_Category, Added Performance_Category, Renamed Columns, Added Overall_Satisfaction, Added Attrition_Risk, Replaced Value, Changed DataType, Trimmed Text, Removed Constant value Column, Reordered Columns1, and Removed Duplicates. The formula bar at the top shows the formula: 'Table.Distinct(#\"Reordered Columns1\", {\"Employee_ID\"})'.

Figure 1. Supplied Power Query screenshot: Cleaned_Table, 43-column output and Applied Steps sequence.

8. Data Quality Assessment

Quality control	Result	Assessment
Total records	1,470	Complete analytical population
Employee_ID duplicates	0	No duplicate employee identifiers remain
Missing values in Cleaned_Data	0	No nulls observed in cleaned output
Attrition / flag inconsistencies	0	Attrition and Attrition_Flag reconcile
Employee status / attrition	Reconciles	Active Employee = Attrition No; Ex-Employee = Attrition Yes
Raw vs cleaned row count	1,470 → 1,470	No row-count loss; cleaning is record-preserving

Quality Interpretation

The cleaned dataset is structurally strong for the intended analysis. Employee_ID is unique, critical attrition fields are populated, and Attrition_Flag is fully consistent with the Attrition text field. The duplicate-removal step therefore acts as an explicit control rather than a transformation that materially reduces the final population.

9. Power BI Model & Report Architecture

The supplied PBIX contains one report page named 'OVERVIEW'. The page contains 117 visual containers, including KPI cards, charts, slicers, tables, images, shapes and text elements. The report uses a dedicated _Measures layer and dimension-style entities for employee profile, job role and work experience.

Observed report component	Evidence
Report page	OVERVIEW
Visual containers	117
Card visuals	12
Slicers	6
Donut charts	1
Bar chart	1
Combo charts	2
Treemap	1
Table visuals	2

Observed Analytical Entities

Entity / field group	Observed dashboard role
Fact_Employee	Employee-level analytical fields
Dim_Employee_Profile	Age group, gender, marital status, education field
Dim_Income	Income level analysis and salary based employee segmentation
Dim_JobRole	Department and job role
Dim_Travel_Distance	Employee commute distance and Travel impact analysis
Dim_Work_Experience	Company tenure and experience group
_Measures	Reusable KPI and analytical measures

10. Power BI Measures

Measure family	Observed measures
Workforce	TOTAL EMPLOYEES; Active Employees; Attrition Count
Rates	Attrition Rate; % attrition of total employee; % Active Emp from Total; % of Total Employee; ATTRITION RATE
Averages	AVERAGE AGE; Average Tenure; Average Salary; Average Overall Satisfaction

Measure Architecture

- The dedicated _Measures table keeps headline calculations separate from raw employee attributes.
- Attrition Count and workforce counts support the executive KPI cards.
- Rate measures support gender, age, role, department and tenure visuals.
- Average measures support the Average Employee Metrics table.
- Visual queries show the use of fields from Dim_Employee_Profile, Dim_JobRole, Dim_Work_Experience and Fact_Employee.

The PBIX package exposes measure names and visual-query metadata but not every DAX expression as plain text.

Accordingly, this report documents the observed measure architecture rather than inventing unavailable formula bodies.

11. Dashboard Evidence & Interpretation

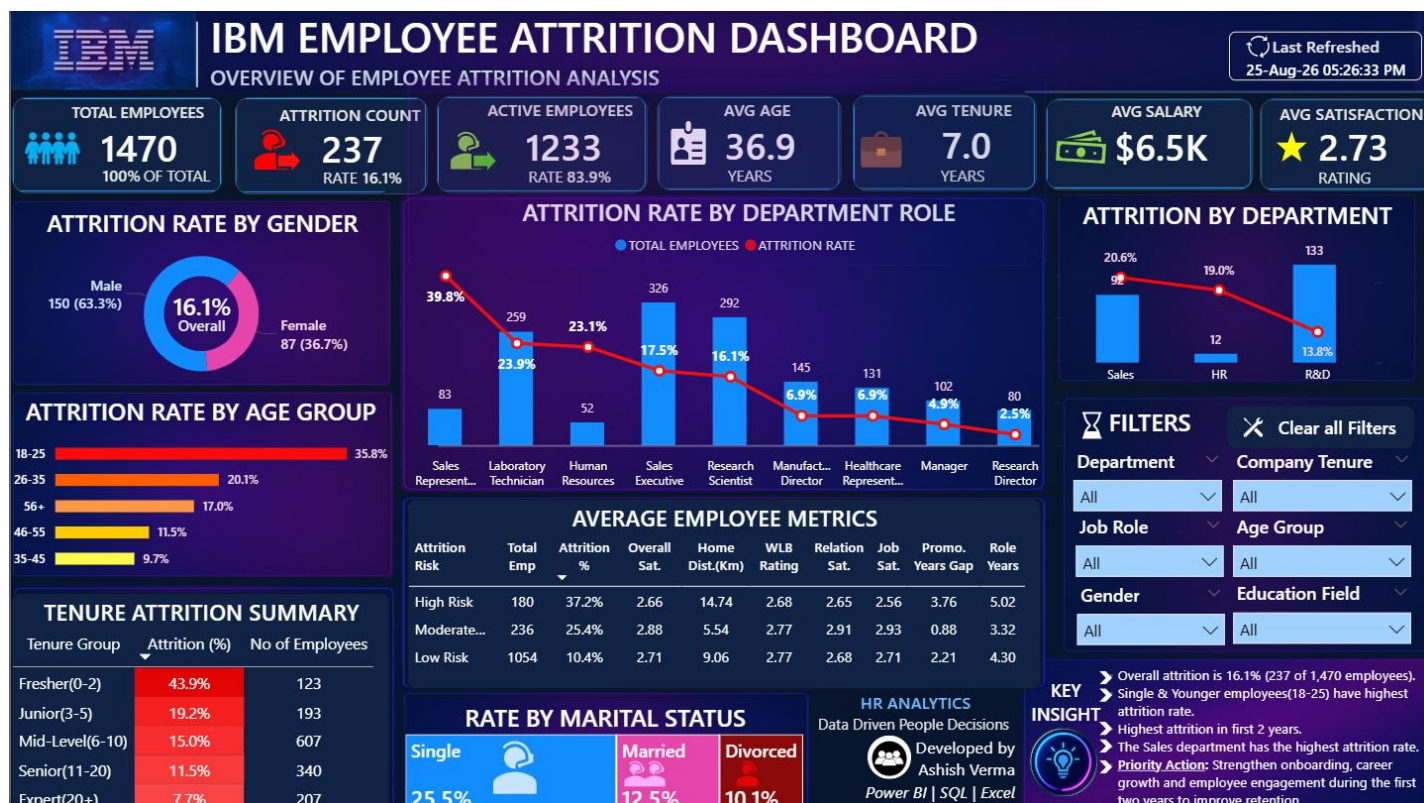


Figure 2. Supplied Power BI dashboard screenshot — IBM Employee Attrition Dashboard.

The dashboard is designed as a single-page executive overview. It combines KPI cards at the top, demographic and tenure visuals on the left, department/job-role analysis in the center, department and filter controls on the right, an Average Employee Metrics table, and a Key Insight panel.

Dashboard Interaction Layer

Slicer	Purpose
Department	Focus on organizational unit
Company Tenure	Focus on tenure stage
Job Role	Focus on role-level retention
Age Group	Focus on employee life stage
Gender	Focus on gender segmentation
Education Field	Focus on educational background

Dashboard Headline Values

Dashboard KPI	Value
Total Employees	1,470
Attrition Count	237
Attrition Rate	16.1%
Active Employees	1,233
Active Rate	83.9%
Average Age	36.9
Average Tenure	7.0 years
Average Salary	\$6.5K
Average Satisfaction	2.73

12. Department & Job Role Analysis

Department	Employees	Left	Attrition Rate	Avg Monthly Income
Sales	446	92	20.6%	\$7.0K
Human Resources	63	12	19.0%	\$6.7K
Research & Development	961	133	13.8%	\$6.3K

department	Total_Employees	Employees_Left	Attrition_Rate	Active_Employees	Emp_Active_Rate	Average_Monthly_Income
Sales	446	92	20.63	354	79.37	6959.2
Human Resources	63	12	19.05	51	80.95	6654.5
Research & Development	961	133	13.84	828	86.16	6281.3

Result 3 x

Output

Action Output

#	Time	Action	Message
16	11:22:20	SELECT department, COUNT(*) AS Total_Employees, SUM(CASE WHEN Attrition='Yes' THEN 1 E...	3 row(s) returned

Figure 3. Independent SQL Validation- attrition rate by department.

Job Role Analysis

Job Role	Employees	Left	Attrition Rate
Sales Representative	83	33	39.8%
Laboratory Technician	259	62	23.9%
Human Resources	52	12	23.1%
Sales Executive	326	57	17.5%
Research Scientist	292	47	16.1%
Manufacturing Director	145	10	6.9%
Healthcare Representative	131	9	6.9%
Manager	102	5	4.9%
Research Director	80	2	2.5%

job_role	Total_Employees	Employees_Left	Attrition_Rate	Active_Employees	Emp_Active_Rate	Avg_Salary	Total_Monthly_Income
Sales Representative	83	33	39.76	50	60.24	2626	217958
Laboratory Technician	259	62	23.94	197	76.06	3237	838427
Human Resources	52	12	23.08	40	76.92	4236	220259
Sales Executive	326	57	17.48	269	82.52	6924	2257315
Research Scientist	292	47	16.10	245	83.90	3240	946072

Result 5 x

Output

Action Output

#	Time	Action	Message
18	11:23:51	SELECT job_role, COUNT(*) AS Total_Employees, SUM(CASE WHEN Attrition='Yes' THEN 1 E...	9 row(s) returned

Figure 4. Independent SQL Validation — highest job-role attrition rates.

Sales has the highest department attrition rate at 20.6%, followed by Human Resources at 19.0%. At role level, Sales Representative is the highest at 39.8%, followed by Laboratory Technician at 23.9%.

13. Demographic & Age Analysis

	Age_Group	Total_Emp	Total_Emp_Left	Emp_Left_Rate	Total_Active_Emp	Emp_Active_Rate
▶	18-25	123	44	35.77	79	64.23
	26-35	528	106	20.08	422	79.92
	56+	47	8	17.02	39	82.98
	46-55	226	26	11.50	200	88.50
	35-45	546	53	9.71	493	90.29

Result 7 ×

Output

Action Output

#	Time	Action	Message
✓ 20	11:26:46	WITH Emp_Details AS (SELECT employee_id,age,attrition, CASE WHEN age >= 56 THEN '...	5 row(s) returned

Figure 5. Independent SQL Validation — Age Group Analysis

Age Group

Age Group	Employees	Left	Attrition Rate
18-25	123	44	35.8%
26-35	528	106	20.1%
56+	47	8	17.0%
46-55	226	26	11.5%
35-45	546	53	9.7%

Gender

Gender	Employees	Left	Attrition Rate
Male	882	150	17.0%
Female	588	87	14.8%

Marital Status

Marital Status	Employees	Left	Attrition Rate
Single	470	120	25.5%
Married	673	84	12.5%
Divorced	327	33	10.1%

The strongest age-related retention signal is among 18–25-year-olds, where attrition reaches 35.8%. Employees aged 35–45 have the lowest rate at 9.7%. Male attrition is 17.0% versus 14.8% for female employees. Single employees have the highest marital-status attrition at 25.5%.

14. Experience & Tenure Analysis

Experience Group	Employees	Left	Attrition Rate
Fresher(0-2)	123	54	43.9%
Junior(3-5)	193	37	19.2%
Mid-Level(6-10)	607	91	15.0%
Senior(11-20)	340	39	11.5%
Expert(20+)	207	16	7.7%

Company Tenure

Company Tenure	Employees	Left	Attrition Rate
New Joiner	44	16	36.4%
1-3 Years	426	106	24.9%
4-7 Years	472	60	12.7%
8-10 Years	282	35	12.4%
10-15 Years	108	7	6.5%
15+ Years	138	13	9.4%

Attrition decreases substantially with career experience: Fresher employees show 43.9%, while Expert employees show 7.7%. Company tenure shows a similar pattern, with New Joiners at 36.4% and employees with 10–15 years at 6.5%.

15. Compensation & Performance Analysis

Income Category	Employees	Left	Attrition Rate	Avg Monthly Income
Lower Income	395	113	28.6%	\$2.4K
Upper Middle Income	148	20	13.5%	\$11.9K
Lower Middle Income	519	66	12.7%	\$4.5K
Middle Income	275	33	12.0%	\$7.7K
Higher Income	133	5	3.8%	\$18.1K

Interpretation

- Lower Income employees have the highest attrition rate at 28.6%.
- Higher Income employees have the lowest attrition rate at 3.8%.
- The Lower Income group averages about \$2.4K monthly income, while Higher Income averages about \$18.1K.
- Performance_Category contains two groups: Good (1,244 employees) and Excellent (226). Their attrition rates are 16.1% and 16.4%, respectively.

16. Satisfaction, Overtime & Work Conditions

Overtime	Employees	Left	Attrition Rate	Avg Overall Satisfaction
Yes	416	127	30.5%	2.79
No	1054	110	10.4%	2.71

Business Travel

Business Travel	Employees	Left	Attrition Rate	Avg Distance (km)
Travel Frequently	277	69	24.9%	9.28
Travel Rarely	1043	156	15.0%	9.09
Non-Travel	150	12	8.0%	9.76

Work-Life Balance

Work-Life Balance Rating	Employees	Left	Attrition Rate
1	80	25	31.2%
2	344	58	16.9%
3	893	127	14.2%
4	153	27	17.6%

Overtime is the clearest work-condition signal in the supplied data: attrition is 30.5% among overtime employees versus 10.4% among non-overtime employees. Travel Frequently also has a higher attrition rate (24.9%) than Non-Travel (8.0%).

17. Attrition Risk & Driver Analysis

Risk Category	Employees	Left	Attrition Rate	Avg Satisfaction	Avg Distance	Avg Job Satisfaction
High Risk	180	67	37.2%	2.66	14.74	2.56
Moderate Risk	236	60	25.4%	2.88	5.54	2.93
Low Risk	1054	110	10.4%	2.71	9.06	2.71

Risk Rule Used in the Project

The supplied SQL implements the same three-level risk logic represented in the cleaned data: employees working without overtime are Low Risk; among overtime employees, low overall satisfaction contributes to High Risk, while long promotion gaps or long distance from home also trigger High Risk; remaining employees are Moderate Risk. The SQL uses the condition structure with OR logic, and the resulting classification exactly reconciles to the Attrition_Risk field in Cleaned_Data.

Risk Findings

- High Risk contains 180 employees and has a 37.2% attrition rate.
- Moderate Risk contains 236 employees and has a 25.4% attrition rate.
- Low Risk contains 1,054 employees and has a 10.4% attrition rate.
- The High Risk group averages 3.76 years since promotion and 14.74 km from home.

18. SQL Validation & Analysis

Section	SQL analysis scope
01	Database & Table Setup
02	Data Import & MySQL Configuration
03	Data Quality & Structure Validation
04	Query Performance Optimization
05	Overall HR KPI Analysis
06	Department & Job Role Analysis
07	Demographic & Age Analysis
08	Experience & Tenure Analysis
09	Compensation & Performance Analysis
10	Employee Satisfaction & Work Conditions
11	Attrition Driver & Business Analysis

SQL Techniques Demonstrated

- CREATE DATABASE and CREATE TABLE for a structured analytical schema.
- LOAD DATA INFILE for bulk import of the cleaned dataset.
- COUNT, SUM, AVG, ROUND and CASE for KPI construction.
- GROUP BY and ORDER BY for segment-level analysis.
- Window functions using SUM(COUNT(*)) OVER() for percentage-of-total calculations.
- CTEs and ROW_NUMBER() for top-3 job roles within each department.
- Conditional aggregation for attrition and active-employee counts.
- CREATE INDEX for common filtering/grouping columns.
- Single-table analytical design without unnecessary JOIN operations.

SQL Quality Checks

- SHOW CREATE TABLE validates the created schema.
- LIMIT 10 provides a sample data inspection.
- COUNT(*) validates row population.
- Duplicate Employee_ID detection validates key uniqueness.
- Missing Employee_ID and Attrition checks validate critical fields.
- Attrition_Flag reconciliation validates derived status consistency.

19. Cross-Layer Reconciliation

Metric	Excel/Cleaned_Data	Dashboard evidence	SQL logic
Total Employees	1,470	1,470	COUNT(*)
Employees Left	237	237	SUM(CASE Attrition='Yes')
Active Employees	1,233	1,233	SUM(CASE Attrition='No')
Attrition Rate	16.1%	16.1%	Employees Left ÷ COUNT(*)
Active Rate	83.9%	83.9%	Active Employees ÷ COUNT(*)
Average Age	36.9	36.9	AVG(Age)
Average Tenure	7.0	7.0	AVG(Years_At_Company)
Average Salary	\$6.5K	\$6.5K	AVG(Monthly_Income)
Average Satisfaction	2.73	2.73	AVG(Overall_Satisfaction)

Validation Result

The headline workforce KPIs reconcile across the cleaned Excel table, the Power BI dashboard screenshot and the SQL definitions. The cleaned data also shows zero duplicate Employee_ID values, zero missing values in Cleaned_Data and zero Attrition/Attrition_Flag inconsistencies.

20. Key Business Insights

- 1.** Overall attrition is 16.1%: 237 of 1,470 employees left.
- 2.** Sales is the highest-attrition department at 20.6%.
- 3.** Sales Representative is the highest-attrition job role at 39.8%.
- 4.** Employees aged 18–25 have a 35.8% attrition rate, the highest age-group rate.
- 5.** Fresher employees (0–2 years of total experience) have the highest experience-group attrition at 43.9%.
- 6.** New Joiners have 36.4% attrition, while employees with 10–15 years at the company have 6.5%.
- 7.** Overtime employees have 30.5% attrition versus 10.4% without overtime.
- 8.** Travel Frequently has 24.9% attrition versus 8.0% for Non-Travel.
- 9.** Single employees have 25.5% attrition, higher than Married and Divorced employees.
- 10.** Lower Income employees have 28.6% attrition, while Higher Income employees have 3.8%.
- 11.** High Risk employees show 37.2% attrition, over three times the Low Risk rate of 10.4%.
- 12.** Human Resources has a small workforce base but a relatively high attrition rate of 19.0%, warranting monitoring alongside Sales.

21. Recommendations

Recommendation	Rationale from analysis	Priority
Strengthen early-career retention	18–25, Fresher and New Joiner groups show the highest attrition.	High
Review overtime workload	Overtime attrition is nearly three times the non-overtime rate.	High
Target Sales retention	Sales has the highest department attrition and Sales Representative is the highest-risk role.	High
Build role-specific retention plans	Laboratory Technician and HR roles also show elevated attrition.	High
Monitor High Risk employees	High Risk attrition is 37.2% versus 10.4% for Low Risk.	High
Improve first-2-year onboarding	Early company tenure is associated with materially higher attrition.	High
Review travel burden	Frequent travelers have higher attrition than non-travel employees.	Medium
Use compensation segmentation	Lower Income employees show the highest attrition; combine pay review with role and tenure analysis.	Medium
Track satisfaction alongside workload	Overall and job satisfaction are included in the risk framework and should be monitored with overtime and promotion gaps.	Medium
Maintain SQL QA checks	Re-run identifier, missing-field and attrition-flag validation after dataset refreshes.	High

22. Limitations & Analytical Controls

Limitations

- The dataset is employee-level and does not contain a time-series employee history, so the analysis describes observed attrition patterns rather than monthly attrition movement.
- Attrition relationships are descriptive and should not be interpreted as proof of causation.
- The derived Attrition_Risk classification is a project-defined rule, not a validated predictive model.
- Employee_ID is the analytical key; no separate historical employee-event table is provided for longitudinal retention analysis.
- The Power BI package exposes measure names and visual metadata, but the supplied artifacts do not expose every DAX formula in plain text.

Controls

- Duplicate Employee_ID check in SQL.
- Missing critical-field validation.
- Attrition versus Attrition_Flag consistency validation.
- Power Query duplicate-removal step.
- Cross-layer KPI reconciliation between Excel, Power BI evidence and SQL.
- Use of explicit derived categories to standardize segmentation.

23. Conclusion

The IBM Employee Attrition Analysis project provides a complete HR analytics workflow for a workforce of 1,470 employees. The solution transforms the source dataset into a 43-column analytical table, applies repeatable Power Query transformations, delivers a Power BI executive dashboard and validates the core workforce metrics through MySQL.

The central business message is clear: attrition is most concentrated among early-career and younger employees, overtime employees, selected sales and technical roles, and the project's High Risk segment. With an overall attrition rate of 16.1%, the strongest retention opportunity is to focus on the first years of employment, workload/overtime, role-specific retention and proactive monitoring of High Risk employees.

The project is particularly portfolio-ready because it demonstrates a complete chain from data preparation to business decision support, while maintaining an independent SQL validation layer and explicit data-quality controls.

24. Technical Skills Demonstrated

Skill	Project evidence
Excel	Raw and cleaned employee worksheets; structured analytical table
Power Query	23 observed Applied Steps covering typing, grouping, replacements, derived fields and duplicate removal
Data Cleaning	Text trimming, value replacement, standardization and constant-column removal
Feature Engineering	Age, income, tenure, experience, distance, performance, satisfaction and risk categories
Power BI Modeling	Fact_Employee plus dimensional entities and dedicated _Measures layer
DAX / Measures	Workforce counts, attrition rates and average employee metrics
Dashboard Design	KPI cards, donut charts, combo chart, bar chart, treemap, table and six slicers
SQL	Schema creation, bulk import, validation, conditional aggregation, CTE, window functions and indexes
HR Analytics	Attrition, retention, demographics, compensation, satisfaction and work-condition analysis
Business Analysis	Key findings, risk prioritization and retention recommendations
Validation	Cross-layer reconciliation of headline KPIs

Appendix A — Data Dictionary

Field	Definition	Role
Employee_ID	Unique employee identifier	Identifier
Employee_Status	Active Employee / Ex-Employee	Derived status
Attrition_Flag	1 for Yes, 0 for No	Derived flag
Attrition	Employee attrition indicator	Outcome
Age	Employee age	Demographic
Age_Group	Age band	Derived category
Gender	Employee gender	Demographic
Marital_Status	Marital status	Demographic
Education	Education level code	Education
Education_Field	Education specialization	Education
Department	Organizational department	Organization
Job_Role	Employee job role	Organization
Job_Level	Job level	Organization
Job_Involvement	Job involvement rating	Engagement
Business_Travel	Travel frequency	Work condition
Distance_From_Home	Distance from home	Work condition
Distance_Category	Distance band	Derived category
Daily_Rate	Daily pay-related rate	Compensation
Hourly_Rate	Hourly pay-related rate	Compensation
Monthly_Rate	Monthly pay-related rate	Compensation
Monthly_Income	Monthly income	Compensation
Income_Range	Monthly income band	Derived category
Income_Category	Income classification	Derived category
Percent_Salary_Hike	Salary increase percentage	Compensation
Stock_Option_Level	Stock option level	Compensation
Performance_Rating	Performance rating	Performance
Performance_Category	Good / Excellent classification	Derived category
Environment_Satisfaction	Environment satisfaction rating	Satisfaction
Job_Satisfaction	Job satisfaction rating	Satisfaction
Relationship_Satisfaction	Relationship satisfaction rating	Satisfaction
Work_Life_Balance	Work-life balance rating	Satisfaction
Overall_Satisfaction	Composite satisfaction measure	Derived metric
Over_Time	Overtime indicator	Work condition
Attrition_Risk	Low / Moderate / High risk classification	Derived risk
Total_Working_Years	Total work experience	Career
Experience_Group	Career experience band	Derived category
Years_At_Company	Years at company	Tenure
Company_Tenure	Company tenure band	Derived category
Years_In_Current_Role	Years in current role	Career
Years_Since_Last_Promotion	Years since promotion	Career
Years_With_Current_Manager	Years with current manager	Career
Num_Companies_Worked	Number of prior companies	Career
Training_Times_Last_Year	Training count in last year	Development

Appendix B — SQL Query Inventory

SQL section	Scope
01. Database & Table Setup	Create hr_analytics database and employee_attrition table.
02. Data Import & MySQL Configuration	Enable local_infile and load cleaned CSV.
03. Data Quality & Structure Validation	Schema preview, sample rows, row count, duplicate key, missing fields and attrition-flag consistency.
04. Query Performance Optimization	Indexes on Attrition, Department and Job_Role.
05. Overall HR KPI Analysis	Workforce size, attrition, active population, averages and distribution.
06. Department & Job Role Analysis	Department/role attrition and compensation; top 3 roles within each department.
07. Demographic & Age Analysis	Age groups, gender and marital status.
08. Experience & Tenure Analysis	Total work experience and company tenure.
09. Compensation & Performance Analysis	Income category, attrition and average monthly income.
10. Employee Satisfaction & Work Conditions	Overtime, satisfaction, travel and work-life balance.
11. Attrition Driver & Business Analysis	Multi-factor High/Moderate/Low risk exposure and supporting averages.

Appendix C — Evidence Register

Artifact	Evidence captured in this report
Employee_Attrition_Cleaned_Data.xlsx	Raw worksheet, cleaned Cleaned_Data, field structure and analytical values
IBM_Employee_Attrition_Analysis.pbix	OVERVIEW page architecture, visual metadata, measures and model field references
IBM_Employee_Attrition_SQL_Validation.sql	Database schema, import, validation, KPI, role, demographic, tenure, compensation, satisfaction and risk analysis
01_Dashboard_Overview.png.png	Power BI IBM Employee Attrition Dashboard screenshot
.1_Power_Query_Applied_Steps.png	Power Query Cleaned_Table screenshot and Applied Steps evidence
Independent analytical charts	Derived directly from supplied Cleaned_Data values for experience, overtime, department and role analysis

Final Deliverable Note

This document is designed as the end-to-end professional report accompanying the supplied Excel workbook, PBIX dashboard and MySQL SQL script. It preserves the project's own terminology and evidence while presenting the work as a coherent portfolio case study.